Solution to Exercise 06 — Analysis on Manifolds, 80416

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Question 1

Subquestion 1

Let $M \subseteq \mathbb{R}^d$ be a C^r manfiold of dimension m, let $p \in M$ and let $\varphi_i : U_i \to M$ for $i \in \{1, 2\}$ be two local parameterizations around p. We will show that $Im\,(D\varphi_1)_{\varphi_1^{-1}(p)} = Im\,(D\varphi_2)_{\varphi_2^{-1}(p)}$.

Proof: Let

$$u_1 = \varphi_1^{-1}(p) \quad u_2 = \varphi_2^{-1}(p)$$

Since φ_1, φ_2 are local parameterizations around p the transition map $\tau = \varphi_2^{-1} \circ \varphi_1$ is well defined and of class C^r on a neighborhood of u_1 and $\varphi_1 = \varphi_2 \circ \tau$.
From the chain rule at u_1 is get that

$$(D\varphi_1)|_{u_1} = (D\varphi_2)|_{u_2} \circ (D\tau)|_{u_1}$$

So for each $v \in \mathbb{R}^m$ we have

$$(D\varphi_1)|_{u_1}(v) = (D\varphi_2)|_{u_2}\Big((D\tau)|_{u_1}(v)\Big)$$

Hence every vector in $Im\Big((D\varphi_1)|_{u_1}\Big)$ is in $Im\Big((D\varphi_2)|_{u_2}\Big)$ which means $Im\Big((D\varphi_1)|_{u_1}\Big) \subseteq Im\Big((D\varphi_2)|_{u_2}\Big)$.
If we reverse the roles of φ_1 and φ_2 we get the $\supseteq$ hence $Im\,(D\varphi_1)_{\varphi_1^{-1}(p)} = Im\,(D\varphi_2)_{\varphi_2^{-1}(p)}$. □

Subquestion 2

We will conclude from section a that if $M, N \subseteq \mathbb{R}^d$ are two manifolds of dimensions m and n respectively then M intersect N transversally at $p \in M \cap N$ if and only if there exist two local parameterizations around $p \in M \cap N$, $\varphi : U \to \mathbb{R}^d, \psi : V \to \mathbb{R}^d$ in M and N respectively such that $Im\,(D\varphi)_{\varphi^{-1}(p)} + Im\,(D\psi)_{\psi^{-1}(p)} = \mathbb{R}^d$.

Proof: $\Longrightarrow$ Assume that M intersect N transversally at $p \in M \cap N$.
This direction is purly from the definition we gave in the recitation: we defined as the following Let $M, N \subseteq \mathbb{R}^d$ be two manifold of dimensions m and n respectively. We say that M intersects N transversally at $p \in M \cap N$ if every two local parameterizations around p, $\varphi : U \to \mathbb{R}^d, \psi : V \to \mathbb{R}^d$ where $U \subseteq \mathbb{R}^m, V \subseteq \mathbb{R}^n$ are open sets, satisfy

$$Im(D\varphi)_{\varphi^{-1}(p)} + Im(D\psi)_{\psi^{-1}(p)} = \mathbb{R}^d$$

And we saied that M intersects N transversally and we denote $M \pitchfork N$ if they intersect transversally at every $p \in M \cap N$.
So this direction is purly from definition.
$\Longleftarrow$ Assume that there exist local parametrizations $\varphi : U \to \mathbb{R}^d, \psi : V \to \mathbb{R}^d$ such that

$$Im\Big((D\varphi)|_{\varphi^{-1}(p)}\Big) + Im\Big((D\psi)|_{\psi^{-1}(p)}\Big) = \mathbb{R}^d$$

From the previous section, we proved that the image of the differential of a local parametrization at p is strictly independent of the parametrization chosen. Therefore, these images uniquely define the tangent spaces

$$T_pM = Im\,(D\varphi)_{\varphi^{-1}(p)} \quad \text{and} \quad T_pN = Im\,(D\psi)_{\psi^{-1}(p)}$$

So from our assumption we get

$$T_pM + T_pN = \mathbb{R}^d$$

This is exactly the definition of M and N intersecting transversally at p. □

Question 2

Subquestion 1

Let $f : \mathbb{R} \to \mathbb{R}$ be a smooth function and let $p \in \Gamma(f) = \{(x, f(x)) : x \in R\} \subseteq \mathbb{R}^2$.
For every v in $\mathbb{R}^2$ we consider the set $L_v = \{p + tv : t \in \mathbb{R}\}$.
We will show that L_v is a manifold and find all L_v such that L_v and $\Gamma(f)$ intersect transversally at p.

Proof: Let $p = (x_0, f(x_0)) \in \Gamma(f)$ and let $v = (v_1, v_2) \in \mathbb{R}^2$. If $v = (0, 0)$ then $L_v = \{p\}$ which is a single point and trivially a 0-dimensional manifold.
Otherwise, assume $v \neq (0, 0)$ so we can define a global parametrization $\varphi : \mathbb{R} \to \mathbb{R}^2$ by $\varphi(t) = p + tv$.
We need to show that this is a local parametrization so we need to show that the domain is open, verify the image, continuity and smoothness and show that it has full rank differential.
Indeed, the domain is $\mathbb{R}$ which is open and $Im(\varphi) = L_v$ and φ is an affine transformation which is C^∞.
We have that φ is a continuous bijection onto L_v and its inverse is (up to a scalar) the dot product with v which is continuous thus φ is a homeomorphism onto its image.
Last, we have $D\varphi_t = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ and since $v \neq 0$ this has full rank hence the differential is injective.
So, L_v is indeed a 1-dimensional C^∞ manifold for $v \neq 0$.
We need to find all L_v such that L_v intersects $\Gamma(f)$ transversally at p.
The graph $\Gamma(f)$ has a global parametrization $\alpha(x) = (x, f(x))$. Its differential at x_0 is the column vector $D\alpha_{x_0} = \begin{pmatrix} 1 \\ f'(x_0) \end{pmatrix}$. Hence

$$T_p\Gamma(f) = \operatorname{Span}\{(1, f'(x_0))\}$$

If $v = 0$, then L_v is a 0-manifold and $T_pL_0 = \{0\}$. The sum of the tangent spaces would just be $T_p\Gamma(f)$, which is 1-dimensional and cannot span $\mathbb{R}^2$. Thus, we must have $v \neq 0$.
If $v \neq 0$, using the parametrization $\varphi(t) = p + tv$, the differential at $t = 0$ is exactly the direction vector v. Thus

$$T_pL_v = \operatorname{Span}\{v\} = \operatorname{Span}\{(v_1, v_2)\}$$

For the sum of these two 1-dimensional subspaces to be the entire 2-dimensional space $\mathbb{R}^2$, their spanning vectors must be linearly independent. Two vectors in $\mathbb{R}^2$ are linearly independent if and only if the determinant of the matrix they form is non-zero

$$\det\begin{pmatrix} 1 & v_1 \\ f'(x_0) & v_2 \end{pmatrix} \neq 0 \Longleftrightarrow 1 \cdot v_2 - v_1 \cdot f'(x_0) \neq 0 \Longleftrightarrow v_2 \neq v_1 f'(x_0)$$

Therefore, L_v and $\Gamma(f)$ intersect transversally at p if and only if $v \neq 0$ and v is not a scalar multiple of $(1, f'(x_0))$. □

Subquestion 2

Let $S^n = \{x \in \mathbb{R}^{n+1} : \|x\| = 1\}$ and for $t \in R$ let $H_t = \{x \in \mathbb{R}^{n+1} : x_{n+1} = t\}$. We will prove that H_t is a manifold and find all $t \in R$ such that $H_t \pitchfork S^n$.

Proof: H_t is a manifold because if we look at $f : \mathbb{R}^{n+1} \to \mathbb{R}$ defined by $f(x) = x_{n+1} - t$ we get a C^∞ function and its differential is the constant $1 \times (n + 1)$ matrix $Df_x = \begin{pmatrix} 0 & \dots & 0 & 1 \end{pmatrix}$ with a full rank at all points $x \in \mathbb{R}^{n+1}$.
By the definition of manifold as the null set of a system of equations we saw in the recitation, the pre-image $f^{-1}(0) = H_t$ is a smooth manifold of dimension $(n + 1) - 1 = n$.
By definition, H_t and S^n intersect transversally if for every $p \in H_t \cap S^n$, their tangent spaces span the ambient space

$$T_pH_t + T_pS^n = \mathbb{R}^{n+1}$$

Let $p = (p_1, ..., p_n, t) \in H_t \cap S^n$, from the definition of H_t its tangent space is $T_pH_t \coloneqq \{v = (v_1, ..., v_{n+1}) \mid v_{n+1} = 0\} = (\mathrm{e}_{n+1})^\perp$.
We consider $g(x) = \|x\|^2$ then $Dg_p(v) = 2\langle p, v\rangle$ hence $T_pS^n = \ker Dg_p = \{v \mid \langle p, v\rangle = 0\} = p^\perp$.
By Rank-Nullity we obtain that T_pH_t and T_pS^n are n-dimensional hyperplanes in $\mathbb{R}^{n+1}$ since $\dim(T_pH_t) = n = \dim(T_pS^n)$.
Because both tangent spaces are defined by the kernel of a $1 \times (n + 1)$ matrix with rank 1, they both lose exactly 1 dimension from the ambient space $\mathbb{R}^{n+1}$, making them both n-dimensional and both with codim of 1.
Therefore $T_pH_t + T_pS^n = \mathbb{R}^{n+1}$ if and only if they are distinct subspaces and they are identical if and only if their normal vectors are linearly dependent.
Since the normals are respectively e_{n+1}, p the sum fails to be all of $\mathbb{R}^{n+1}$ exactly when these normals are parallel which means $p = \lambda\mathrm{e}_{n+1}$.
Since $p \in S^n$ we have $p = \pm\mathrm{e}_{n+1}$ but sinch $p \in H_t$ we have $t = \pm 1$ thus $H_t \pitchfork S^n$ if and only if $|t| < 1$ (for $t > 1$ the intersection is empty and when $|t| = 1$ the intersection consists of exactly one point (the pole): $p = (0, ..., 0, \pm 1)$ which is parallel). □

Question 3

Subquestion 1

Let $M, N \subseteq \mathbb{R}^d$ be C^r manifolds of dimensions m and n respectively. We will prove that $M \times N \subseteq \mathbb{R}^d \times \mathbb{R}^d = \mathbb{R}^{2d}$ is a C^r manifold of dimension $m + n$.

Proof: Let $(p, q) \in M \times N$.
Because M is a C^r manifold of dimension m in $\mathbb{R}^d$, there exists an open set $U \subseteq \mathbb{R}^m$, an open neighborhood $W_1 \subseteq \mathbb{R}^d$ of p, and a C^r local parametrization $\varphi : U \to M \cap W_1$ such that the differential $D\varphi_x$ is injective for all $x \in U$.
Similarly, because N is a C^r manifold of dimension n in $\mathbb{R}^d$, there exists an open set $V \subseteq \mathbb{R}^n$, an open neighborhood $W_2 \subseteq \mathbb{R}^d$ of q, and a C^r local parametrization $\psi : V \to N \cap W_2$ such that the differential $D\psi_y$ is injective for all $y \in V$.
We look at the function $\Phi : U \times V \to \mathbb{R}^{2d}$ defined by

$$\Phi(x, y) = (\varphi(x), \psi(y))$$

We need to verify that Φ is a valid local parametrization for $M \times N$ around (p, q) so we need to show that the domain is open, verify the image, continuity and smoothness and show that it has full rank differential.
We know that $U \times V$ is open set in $\mathbb{R}^{m+n}$ since product of open sets is open.
The image of Φ is $\varphi(U) \times \psi(V) = (M \cap W_1) \times (N \cap W_2) = (M \times N) \cap (W_1 \times W_2)$ and since $W_1 \times W_2$ is open set in $\mathbb{R}^{2d}$ that contains (p, q) Φ maps onto an open neighborhood of (p, q) in the subspace topology of $M \times N$. Furthermore, because the Cartesian product of two homeomorphisms is a homeomorphism, Φ is a homeomorphism onto its image and each of the coordinates of Φ is of class C^r so this is also smooth.
We need to show that it has $D\Phi$ is of full rank, we have

$$D\Phi_{(x,y)} = \begin{pmatrix} D\varphi_x & 0 \\ 0 & D\psi_y \end{pmatrix}$$

But since $D\varphi_x$ has full rank of m and $D\psi_y$ has full rank of n the block matrix $D\Phi_{(x,y)}$ has full column rank $m + n$. Thus, the differential is injective everywhere in $U \times V$.
We constructed a local parametrization around any (p, q) hence $M \times N$ is a C^r manifold of dimension $m + n$. □

Subquestion 2

Denote $\Delta = \{(x, y) \in \mathbb{R}^d \times \mathbb{R}^d : x = y\}$. We will prove that $\Delta \subseteq \mathbb{R}^{2d}$ is a manifold of dimension d.

Proof: We can prove this immediately by observing that the diagonal Δ is simply the graph of a smooth function.
Define the identity function $f : \mathbb{R}^d \to \mathbb{R}^d$ by $f(x) = x$, the domain of f is $\mathbb{R}^d$ (which is open), and f is a smooth C^r function.
By definition, the graph of f is:

$$\Delta = \Gamma(f) = \{(x, f(x)) : x \in \mathbb{R}^d\} = \{(x, x) : x \in \mathbb{R}^d\}$$

We saw at exercise 5 that the graph of any C^r function $f : U \subset \mathbb{R}^d \to \mathbb{R}^k$ is a C^r manifold of dimension d. Therefore, Δ is a C^r manifold of dimension d. □

Subquestion 3

We will prove that $M \pitchfork N$ if and only if $M \times N \pitchfork \Delta$.

Proof: Since $\Delta = \{(x, x)\}$ we have $T_{(p,p)} = \{(v, v) \mid v \in \mathbb{R}^d\}$.
From the first section we have $T_{(p,p)}(M \times N) = T_p M \times T_p N$ thus $T_{(p,p)}(M \times N) = \{(u, w) \mid u \in T_p M, w \in T_p N\}$.
$\Longrightarrow$ Assume that $M \pitchfork N$ so $T_p M + T_p N = \mathbb{R}^d$ and we prove $T_{(p,p)}(M \times N) + T_{(p,p)}\Delta = \mathbb{R}^{2d}$.
Let $(a, b) \in \mathbb{R}^{2d}$ and since $a - b \in \mathbb{R}^d = T_p M + T_p N$ there exists $u \in T_p M, w \in T_p N$ such that $a - b = v - w$.
Let $v = a - u = b - w$ then $(a, b) = (u, w) + (v, v)$ but since $(u, w) \in T_{(p,p)}(M \times N)$ and $(v, v) \in T_{(p,p)}\Delta$ we obtain that $(a, b) \in T_{(p,p)}(M \times N) + T_{(p,p)}\Delta$ hence $T_{(p,p)}(M \times N) + T_{(p,p)}\Delta = \mathbb{R}^{2d}$ therefore $M \times N \pitchfork \Delta$.
$\Longleftarrow$ Assume that $T_{(p,p)}(M \times N) + T_{(p,p)}\Delta = \mathbb{R}^{2d}$.
We take any $z \in \mathbb{R}^d$ then $(z, 0) \in \mathbb{R}^{2d}$ and by our assumption there exists $u \in T_p M, w \in T_p N, v \in \mathbb{R}^d$ such that $(z, 0) = (u, w) + (v, v)$ which means $z = u + v$ and $0 = w + v$ thus $v = -w$ and therefore $z = u - w$.
Since $u \in T_p M, -w \in T_p N$ we obtain that $z \in T_p M + T_p N$ and as z was arbitrary we have $T_p M + T_p N = \mathbb{R}^d$ hence $M \pitchfork N$. □